

Normal operators of double fibration transforms

Hiroyuki Chihara (University of the Ryukyus, Okinawa Island, Japan)

19 June 2026, 10th Pacific Rim Conference on Mathematics, Academia Sinica

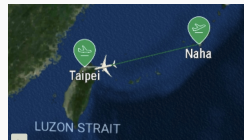


Table of Contents

Double fibration transforms

Z-conjugate points

Structure of normal operators

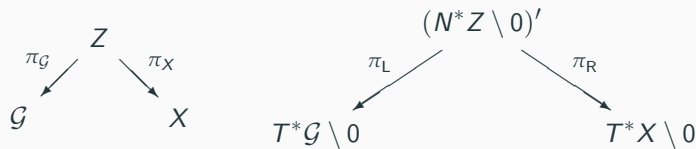
Double fibration transforms



Double fibration

Following Mazzucchelli-Salo-Tzou [3], we introduce double fibration transforms.

- Let $\mathcal{G}$ and X be oriented smooth manifolds without boundaries. $N := \dim(\mathcal{G})$ and $n := \dim(X)$. Denote by $d\mathcal{G}$ and dX the orientation forms of $\mathcal{G}$ and X respectively.
- Let Z be an oriented embedded submanifold of $\mathcal{G} \times X$, and let dZ be the orientation form.
- Assume that $N + n > \dim(Z) > N \geq n \geq 2$, and set $n' := \dim(Z) - N$ and $n'' := n - n'$. Then $\dim(Z) = N + n'$, $n = n' + n''$ and $n', n'' = 1, \dots, n - 1$.



- We assume that Z is a **double fibration**, that is, the natural projections $\pi_{\mathcal{G}} : Z \rightarrow \mathcal{G}$ and $\pi_X : Z \rightarrow X$ are submersions respectively.

Orientation forms on $G_z := \pi_X \circ \pi_{\mathcal{G}}^{-1}(z)$ and $H_x := \pi_{\mathcal{G}} \circ \pi_X^{-1}(x)$

- $G_z := \pi_X \circ \pi_{\mathcal{G}}^{-1}(z)$ becomes an n' -dim submanifold of X for any $z \in \mathcal{G}$, and $H_x := \pi_{\mathcal{G}} \circ \pi_X^{-1}(x)$ forms an $(N - n'')$ -dim submanifold of $\mathcal{G}$ for any $x \in X$.
- Fix arbitrary $(z, x) \in Z$, and let $\{v_1, \dots, v_{n'}\}$ and $\{w_1, \dots, w_N\}$ be bases of $T_x X$ and $T_z \mathcal{G}$ respectively such that

$$T_{(z,x)}Z = \text{span}\langle v_1, \dots, v_{n'}, w_1, \dots, w_N \rangle = \text{span}\langle v_1, \dots, v_{n'}, w_1, \dots, w_{N-n''} \rangle.$$

The induced orientation forms dG_z on G_z and dH_x on H_x are given by

$$\begin{aligned} dG_z(d\pi_X(v_1), \dots, d\pi_X(v_{n'})) &:= dZ_{\pi_{\mathcal{G}}^{-1}(z)}(v_1, \dots, v_{n'}) \\ &= \frac{dZ(v_1, \dots, v_{n'}, w_1, \dots, w_N)}{d\mathcal{G}(d\pi_{\mathcal{G}}(w_1), \dots, d\pi_{\mathcal{G}}(w_N))}, \\ dH_x(d\pi_{\mathcal{G}}(w_1), \dots, d\pi_X(w_{N-n''})) &:= dZ_{\pi_X^{-1}(x)}(w_1, \dots, w_{N-n''}) \\ &= \frac{dZ(v_1, \dots, v_{n'}, w_1, \dots, w_{N-n''})}{dX(d\pi_X(v_1), \dots, d\pi_X(v_{n'}))}. \end{aligned}$$

Double fibration transform

Suppose that a weight function $\kappa(z, x) \in C^\infty(\mathcal{G} \times X)$ is nowhere vanishing. A double fibration transform $\mathcal{R}$ associated with the double fibration Z is defined by

$$\mathcal{R}f(z) := \left(\int_{G_z} \kappa(z, x) \frac{f}{|dX|^{1/2}}(x) dG_z(x) \right) |d\mathcal{G}(z)|^{1/2}$$

for $f \in \mathcal{D}(X, \Omega_X^{1/2})$. The adjoint $\mathcal{R}^*$ is given by

$$\mathcal{R}^*u(x) = \left(\int_{H_x} \overline{\kappa(z, x)} \frac{u}{|d\mathcal{G}|^{1/2}}(z) dH_x(z) \right) |dX(x)|^{1/2}$$

for $u \in \mathcal{D}(\mathcal{G}, \Omega_{\mathcal{G}}^{1/2})$. Then we deduce that

$$\mathcal{R} : \mathcal{D}(X, \Omega_X^{1/2}) \rightarrow \mathcal{E}(\mathcal{G}, \Omega_{\mathcal{G}}^{1/2}), \quad \mathcal{R}^* : \mathcal{D}(\mathcal{G}, \Omega_{\mathcal{G}}^{1/2}) \rightarrow \mathcal{E}(X, \Omega_X^{1/2}),$$

are continuous linear mappings, and so are

$$\mathcal{R} : \mathcal{E}'(X, \Omega_X^{1/2}) \rightarrow \mathcal{D}'(\mathcal{G}, \Omega_{\mathcal{G}}^{1/2}), \quad \mathcal{R}^* : \mathcal{E}'(\mathcal{G}, \Omega_{\mathcal{G}}^{1/2}) \rightarrow \mathcal{D}'(X, \Omega_X^{1/2}).$$

More precisely $\mathcal{R}$ and $\mathcal{R}^*$ are elliptic Fourier integral operators.

Mapping properties of double fibration transforms

Theorem 1 ([3, Theorem 2.2] & [Hörmander IV, Theorem 25.2.2])

Suppose that Z is a double fibration with $\dim(Z) = N + n'$. Then $\mathcal{R}$ and $\mathcal{R}^$ are elliptic Fourier integral operators of order $-(N + 2n' - n)/4$ with canonical relations $(N^*Z \setminus 0)'$ and $((N^*Z \setminus 0)^T)'$ respectively. More precisely*

$$\begin{aligned}\mathcal{R} &\in \mathcal{I}^{-(N+2n'-n)/4}(\mathcal{G} \times X, N^*Z \setminus 0; \Omega_{\mathcal{G} \times X}^{1/2}), \\ \mathcal{R}^* &\in \mathcal{I}^{-(N+2n'-n)/4}(X \times \mathcal{G}, (N^*Z \setminus 0)^T; \Omega_{X \times \mathcal{G}}^{1/2}),\end{aligned}$$

where

$$\begin{aligned}N^*Z \setminus 0 &= \{(z, A(z, x)\eta, x, \eta) : (z, x) \in Z, \eta \in N_x^*G_z \setminus \{0\}\} \\ &= \{(z, \zeta, x, B(z, x)\zeta) : (z, x) \in Z, \zeta \in N_z^*H_x \setminus \{0\}\},\end{aligned}$$

$A(z, x) \in \text{Hom}(N_x^*G_z, T_z^*\mathcal{G})$ and $B(z, x) \in \text{Hom}(N_z^*H_x, T_x^*X)$ smoothly depend on $(z, x) \in Z$ respectively.

Examples of double fibration transforms

- **d -plane transform on $\mathbb{R}^n$:** $\mathcal{G} :=$ the set of all d -dimensional planes in $\mathbb{R}^n$, $X := \mathbb{R}^n$.
- **Geodesic X-ray transform:** Let (M, g) be a compact and nontrapping Riemannian manifold with a strictly convex boundary. Denote by $\nu(x)$ the unit outer normal vector at $x \in \partial M$. Then

$$\mathcal{G} := \partial_- SM = \{(x, u) \in SM : x \in \partial M, \langle u, \nu(x) \rangle < 0\}, \quad X := M^{\text{int}}.$$

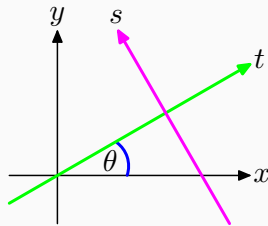
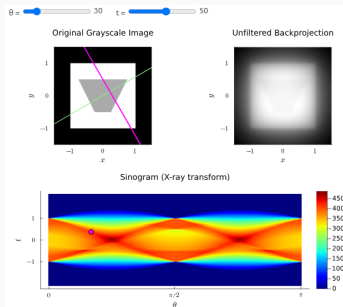
- **Null bicharacteristics:** Let P be a real-principal-type pseudodifferential operator on a manifold X with the principal symbol $p_m(x, \xi)$. Let $\mathcal{G}$ be the set of all Hamilton flows for $p_m(x, \xi) = 0$ on T^*X , and consider the integration over all the $\gamma \in \mathcal{G}$.
- **Light ray transform:** This is a special case of the above. Let (M, g) be a Lorentzian manifold, and set P be the d'Alembertian.

X-ray transform on $\mathbb{R}^2$

Let $n = 2$. The X-ray transform of a function f of $(x, y) \in \mathbb{R}^2$ is defined by

$$R_1 f(\theta, t) := \int_{-\infty}^{\infty} f(t \cos \theta - s \sin \theta, t \sin \theta + s \cos \theta) ds, \quad (\theta, t) \in [0, \pi] \times \mathbb{R}.$$

$R_1^* R_1$ is the normal operator (unfiltered backprojection) of R_1 and $R_1^* R_1 = (-\partial_x^2 - \partial_y^2)^{-1/2}$.
 $(-\partial_x^2 - \partial_y^2)^{1/2} R_1^* R_1 f = f$ is known as the filtered backprojection.



Z-conjugate points

For local coordinates $(z, x) = (z', z'', x', x'') \in \mathbb{R}^{N-n''} \times \mathbb{R}^{n''} \times \mathbb{R}^{n'} \times \mathbb{R}^{n''}$, There exist $\mathbb{R}^{n''}$ -valued functions $\phi(z, x')$ and $b(x, z')$ such that we have locally

$$Z = \{x'' = \phi(z, x')\} = \{z'' = b(x, z')\}.$$

Lemma 2 ([3, Lemmas 2.4, 2.5 and 2.6])

$$N_{(z,x)}^* Z = \left\{ \left(-\phi_z(z, x')^T \eta'', (-\phi_{x'}(z, x')^T \eta'', \eta'') \right) : \eta'' \in \mathbb{R}^{n''} \right\},$$

$$A(z, x) \begin{bmatrix} -\phi_{x'}(z, x')^T \\ I_{n''} \end{bmatrix} \eta'' = -\phi_z(z, x')^T \eta'', \quad \eta'' \in \mathbb{R}^{n''}.$$

Similar results hold for $b(x, z')$ and $B(z, x)$.

Variation fields and conjugate points

Fix arbitrary $(z, w) \in T\mathcal{G}$, we introduce the variation field $J_w : G_z \rightarrow (N_x^* G_z)^*$ is defined by

$$J_w(x) := A(z, x)^* w \simeq -\phi_z(z, x') w \in (N_x^* G_z)^* \simeq N_x G_z = T_x X / T_x G_z$$

for $x \in G_z$. Note that

$$A(z, x)^* \in \text{Hom}(T_z \mathcal{G}, (N_x^* G_z)^*) \simeq \text{Hom}(T_z \mathcal{G}, N_x G_z), \quad (z, x) \in Z.$$

For $z \in \mathcal{G}$ and $x, y \in G_z$, set

$$V_z(x, y) := \{J_w(x) : w \in T_z \mathcal{G}, J_w(y) = 0\}.$$

Note that $\dim(V_z(x, y)) \leq n''$ holds since $\text{rank}(A(z, x)^*) = n''$.

Riemannian manifold case: If $x = \exp_y(tu)$, then $J_w(x) \simeq Y(t) := tD \exp_y(tu)w$.

Definition 3

Suppose that Z is a double fibration and $N \geq 2n''$. Let $k = 1, \dots, n''$.

- **Z-conjugate triplet of degree k :** Let $z \in \mathcal{G}$ and let $x, y \in G_z$ with $x \neq y$. We say that $(z; x, y)$ is a Z -conjugate triplet of degree k if $\dim(V_z(x, y)) = n'' - k$.
- **Regular Z-conjugate triplet of degree k :** We say that a Z -conjugate triplet $(z; x, y)$ of degree k is regular if there exist a nbd U_x of x in X , a nbd U_y of y in X , and a nbd W_z of z in $\mathcal{G}$ such that any Z -conjugate triplet $(z'; x', y') \in W_z \times U_x \times U_y$ is also of degree k . The set of all the regular Z -conjugate triplets of degree k is denoted by $C_{R,k}$.
- The set of all the Z -conjugate triplets which are not regular is denoted by C_S .

How to describe Z -conjugacy 1/2

Let $(z_0; x_0, y_0)$ be a Z -conjugate triplet of degree $k = 1, \dots, n''$. Z is locally expressed by ϕ and ψ as $\{x'' = \phi(z, x')\}$ near (z_0, x_0) and $\{y'' = \psi(z, y')\}$ near (z_0, y_0) respectively, and

$$V_{z_0}(x_0, y_0) = \{\phi_z(z_0, x'_0)w : w \in \text{Ker}(\psi_z(z_0, y'_0))\}.$$

Set $\phi = [\phi^{(1)}, \dots, \phi^{(n'')}]^T$ and $\psi = [\psi^{(1)}, \dots, \psi^{(n'')}]^T$. Then $\{\phi_z^{(1)}, \dots, \phi_z^{(n'')}\}$ and $\{\psi_z^{(1)}, \dots, \psi_z^{(n'')}\}$ are linearly independent near (z_0, x_0) and (z_0, y_0) respectively. Note that

$$\text{Ker}(\psi_z(z_0, y'_0)) = \text{span}\langle \psi_z^{(1)}(z_0, y'_0)^T, \dots, \psi_z^{(n'')}(z_0, y'_0)^T \rangle^\perp \quad \text{in } \mathbb{R}^N.$$

We deduce that $\dim(V_{z_0}(x_0, y_0)) = n'' - k$ is equivalent to

$$\dim\left(\text{span}\langle \phi_z^{(1)}(z_0, x'_0), \dots, \phi_z^{(n'')}(z_0, x'_0) \rangle \cap \text{span}\langle \psi_z^{(1)}(z_0, y'_0), \dots, \psi_z^{(n'')}(z_0, y'_0) \rangle\right) = k. \quad (1)$$

We express (1) by k equations.

How to describe Z -conjugacy 2/2

Denote by $\{\tilde{\psi}_z^{(1)}, \dots, \tilde{\psi}_z^{(n'')}\}$ the Schmidt orthonormalization of $\{\psi_z^{(1)}, \dots, \psi_z^{(n'')}\}$.

(1) is equivalent to the following: there exist $\lambda_1, \dots, \lambda_k \in \mathbb{R}^{n''}$, $\lambda_l = [\lambda_{l1}, \dots, \lambda_{ln''}]$ ($l = 1, \dots, k$) such that $\lambda_1, \dots, \lambda_k$ are linearly independent and if we set

$$\begin{aligned}\phi_z^{\lambda_l}(z, x') &:= \lambda_l \phi_z(z, x') = \sum_{m=1}^{n''} \lambda_{lm} \phi_z^{(m)}(z, x'), \\ H^{\lambda_l}(z, x', y') &:= \phi_z^{\lambda_l}(z, x') \phi_z^{\lambda_l}(z, x')^T - \sum_{m=1}^{n''} |\phi_z^{\lambda_l}(z, x') \tilde{\psi}_z^{(m)}(z, y')^T|^2 \\ &= \lambda_l \phi_z(z, x') (I_N - \tilde{\psi}_z^T(z, y') \tilde{\psi}_z(z, y')) \phi_z(z, x')^T \lambda_l^T, \\ H^\lambda(z, x', y') &:= [H^{\lambda_1}(z, x', y'), \dots, H^{\lambda_k}(z, x', y')]^T,\end{aligned}$$

for $l = 1, \dots, k$, then

$$H^\lambda(z_0, x'_0, y'_0) = 0. \quad (2)$$

An artificial condition (H)

Condition (H): Let $k = 1, \dots, n''$. Suppose that $(z_0; x_0, y_0) \in C_{R,k}$. Suppose that $H^\lambda(z, x', y')$ is the same as that of the previous paragraph and satisfies (2).

Condition (H) is that $\text{rank}(D_{z,x',y'} H^\lambda(z_0, x'_0, y'_0)) = 1$ holds for any choice of linearly independent $\lambda_1, \dots, \lambda_k \in \mathbb{R}^{n''}$.

Spirit of Condition (H):

- **rank one:** We refer the case of the geodesic X-ray transform for the dimension. In other words we fit our case to the geodesic X-ray transform.
- **for any choice of $\lambda_1, \dots, \lambda_k$:** We avoid the case that there exist two choices $\lambda_1, \dots, \lambda_k$ and $\lambda'_1, \dots, \lambda'_k$ such that

$$\{H^\lambda(z, x', y') = 0\} \cap \{H^{\lambda'}(z, x', y') = 0\}$$

is transversal.

Lemma 4

Suppose that Z is a double fibration, $N \geq 2n''$ and some condition (H). For any $k = 1, \dots, n''$, $C_{R,k}$ is an $(N + 2n' - 1)$ -dimensional embedded submanifold of $\mathcal{G} \times X \times X$.

Note that the connected component of $C_{R,k}$ containing $(z_0; x_0, y_0)$ is characterized by

$$F(x'', y'', z; x', y') := \begin{bmatrix} x'' - \phi(z, x') \\ y'' - \psi(z, y') \\ H^\lambda(z, x', y') \end{bmatrix} = 0 \quad \text{near} \quad (z_0; x_0, y_0).$$

We have

$$\text{rank}(DF(x_0'', y_0'', z_0; x_0', y_0')) = \text{rank} \begin{bmatrix} I_{n''} & O & * \\ O & I_{n''} & * \\ O & O & D_{z, x', y'} H^\lambda(z_0, x_0', y_0') \end{bmatrix} = 2n'' + 1.$$

Structure of normal operators

Normal operators without Z -conjugate points

$$\mathcal{R}^* \mathcal{R} f(x) = \left(\iint_{H_x \times G_z} \overline{\kappa(z, x)} \kappa(z, y) \frac{f}{|dX|^{1/2}}(y) dG_z(y) dH_x(z) \right) |dX(x)|^{1/2}, \quad x \in X.$$

Set $\mathcal{C} := (N^*Z \setminus 0)'$, which is the canonical relation of $\mathcal{R}$.

Theorem 5

Suppose that Z is a double fibration. In addition, we assume $N \geq 2n''$ and the following:

- $\pi_X : Z \rightarrow X$ is proper, and $\pi_X^{-1}(x)$ is connected for any $x \in X$.*
- There are no Z -conjugate triplets, and $D\pi_L$ is injective at all $(z, \zeta, x, \eta) \in \mathcal{C}$.*

Then $\mathcal{C}^T \circ \mathcal{C}$ is a clean intersection with excess $e = N - n$, and $\mathcal{R}^ \mathcal{R}$ is an elliptic pseudodifferential operator of order $-n'$ on X .*

Proof: Some lemmas in [3] and the assumptions guarantee the Bolker condition. □

Known results on geodesic X-ray transforms with conjugate points

- Let (M, g) be a compact Riemannian manifold with strictly convex boundary. Set $\gamma_w(t) := \exp_{\pi_M(w)}(tw)$ for $w \in \partial_- SM$. Consider the geodesic X-ray transform

$$\mathcal{X}f(w) := \left(\int_0^{\tau(w)} \kappa(\gamma_w(t), \dot{\gamma}_w(t)) \frac{f}{|dM|^{1/2}}(\gamma_w(t)) dt \right) |d\partial_- SM(w)|^{1/2}.$$

- Stefanov and Uhlmann (2012) [4]: If $v_0 = |v_0|\theta_0$ is a fold conjugate vector at p_0 , and v_0 is the only singularity of $\exp_{p_0}(v)$ on γ_{θ_0} near p_0 , then the localized normal operator is decomposed as

$$\mathcal{X}^* \chi \mathcal{X} = A + F \quad \text{near } p_0,$$

where A is a PsDOs of order -1 , and F is a *FIO* of order $-n/2$.

- Holman and Uhlmann (2018) [2]: If $C_S = \emptyset$, then

$$\mathcal{X}^* \mathcal{X} = A + \sum_{k=1}^{n-1} \sum_{\alpha=1}^{M_k} F_{k,\alpha},$$

where A is a PsDOs of order -1 , and F is a *FIO* of order $-(n - k + 1)/2$.

Normal operators with Z -conjugate points

Theorem 6

Suppose that Z is a double fibration, $C_S = \emptyset$, $N \geq 2n''$, condition (H) and the following:

- π_X is proper, and $\pi_X^{-1}(x)$ is connected for any $x \in X$.
- If $\pi_L^{-1}((z, \zeta)) = \{(z, \zeta, x, \eta)\}$ for $(z, \zeta, x, \eta) \in \mathcal{C}$, then $D\pi_L|_{(z, \zeta, x, \eta)}$ is injective.

Then we have a decomposition of $\mathcal{R}^*\mathcal{R}$ of the form

$$\mathcal{R}^*\mathcal{R} = A + \sum_{k=1}^{n''} \sum_{\alpha \in \Lambda_k} F_{k, \alpha},$$

where A is an elliptic PsDO of order $-n'$ on X ,
 $F_{k, \alpha}$ is a FIO in $\mathcal{I}^{-(n+1-k)/2}(X \times X, \mathcal{C}'_{A_{k, \alpha}}; \Omega_{X \times X}^{1/2})$ with some canonical relation of $\mathcal{C}_{F_{k, \alpha}}$,
associated to the decomposition of connected components $C_{R, k} = \bigcup_{\alpha \in \Lambda_k} C_{R, k, \alpha}$.

Outline of the proof

- $\mathcal{R}^*\mathcal{R}$ is given by

$$\frac{\mathcal{R}^*\mathcal{R}f}{|dX|^{1/2}}(x) = \iint_{H_x \times G_z} \overline{\kappa(z, x)} \kappa(z, y) \frac{f}{|dX|^{1/2}}(y) dG_z(y) dH_x(z).$$

- Follow the idea of Holman and Uhlmann [2]: a partition of unity of $\mathcal{G} \times X \times X$.
- Set $C_\delta := \{(z; x, x) : z \in \mathcal{G}, x \in X\}$, which is related to the elliptic term.
- $C_{R,k,\alpha}$ are disjoint since $C_S = \emptyset$, so are $C_{R,k,\alpha}$ and C_δ .
Pick up disjoint nbds $U_{k,\alpha}$ and U_δ of $C_{R,k,\alpha}$ and C_δ respectively in $\mathcal{G} \times X \times X$.
- We can find an open set U_0 in $\mathcal{G} \times X \times X$ such that

$$U_0 \cup U_\delta \cup (\cup U_{k,\alpha}) = \mathcal{G} \times X \times X, \quad U_0 \cap (C_\delta \cup (\cup C_{R,k,\alpha})) = \emptyset.$$

- Pick up a partition of unity subordinated to $\{U_0, U_\delta, U_{k,\alpha}\}$, and split the Schwartz kernel of $\mathcal{R}^*\mathcal{R}$. U_0 -part of $\mathcal{R}^*\mathcal{R}$ is a smoothing operator, and is absorbed in U_δ -part A .

-  H. Chihara, *Microlocal analysis of double fibration transforms with conjugate points*, J. Geom. Anal., **36** (2026), ID 68, 22 pages.
-  S. Holman and G. Uhlmann, *On the microlocal analysis of the geodesic X-ray transform with conjugate points*, J. Diff. Geom., **108** (2018), pp.459–494.
-  M. Mazzucchelli, M. Salo and L. Tzou, *A general support theorem for analytic double fibration transforms*, arXiv:2306.05906.
-  P. Stefanov and G. Uhlmann, *The geodesic X-ray transform with fold caustics*, Anal. PDE, **5** (2012), pp.219–260.